

Nam Nguyen

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OBJECTIVE

Looking for a Machine Learning & AI position focused on neural data compression (image/video coding), computer vision, generative modeling, and inverse problem - applying advanced information-theoretic frameworks and deep learning to build high-impact intelligent systems.

TECHNICAL SKILLS

- **AI/ML:** Image/Video Compression, Computer Vision, Deep Generative Models, Diffusion Model
- **Quantitative Research:** Mathematical Modeling, Optimization, Statistics & Probability Theory
- **Programming Languages:** Python, MATLAB, C/C++
- **Frameworks/Tools:** PyTorch, TensorFlow, CompressAI, CVX

WORK EXPERIENCE

Communications and Signal Processing Group

Corvallis, Oregon, USA

Ph.D. Research Assistant

Mar. 2022 - Present

Perception-based Image Denoising via Generative Compression

- Designed a perception-driven denoising framework using entropy-coded latent representations to control the rate-distortion-perception tradeoff.
- Developed two conditional generative-compression denoisers using GAN- and diffusion-based reconstruction to improve perceptual image quality.
- Built a PyTorch training and evaluation pipeline and benchmarked on Kodak/DIV2K (synthetic noise) and FMD/SIDD (real noise) using PSNR/SSIM and perceptual metrics.
- **Outputs:** 1 accepted conference paper (ICIP 2026).

Cross-Domain Lossy Compression via Rate- and Classification-constrained Optimal Transport

- Developed a unified compression framework integrating bit-rate, distortion, classification, and perceptual constraints for robust cross-domain generalization.
- Implemented deep compression models (Autoencoder, WGAN) in Python with differentiable quantization and entropy-constrained losses for image restoration tasks (super-resolution, denoising, inpainting).
- Validated on ImageNet and Kodak datasets, showing strong theory-to-practice alignment.
- **Outputs:** 1 accepted conference paper (ICLR 2026) and 1 journal paper manuscript.

Universal Rate-Distortion-Classification (RDC) Representations for Lossy Compression

- Designed a trainable RDC objective coupling rate, distortion, and accuracy to learn semantic representations for compression and downstream inference.
- Built deep compression models in Python with differentiable quantization, entropy- and classification-constrained losses, producing empirical RDC curves.
- Demonstrated the learned latent features on MNIST and SVHN datasets serve as compact codecs supporting classification with minimal accuracy loss versus specialized encoders.
- **Outputs:** 1 accepted conference paper (ITW 2025) and 1 journal paper submission.

Los Alamos National Laboratory

Los Alamos, New Mexico, USA

Machine Learning Research Intern

May 2026 - Aug. 2026

Generative AI for Seismic Imaging and Interpretation

- Develop a novel diffusion-based probabilistic denoising framework to enhance physics-driven full-waveform inversion for seismic imaging and subsurface characterization.
- Design a multimodal learning architecture that integrates heterogeneous geophysical data sources to regularize inverse problems and improve reconstruction accuracy.

- Conduct systematic evaluation of generative models on both synthetic and real-world seismic datasets, analyzing robustness, uncertainty quantification, and imaging performance.
- **Outputs:** 1 journal paper manuscript.

Deakin University Applied Artificial Intelligence Initiative **Burwood, Victoria, Australia**
Machine Learning Research Intern **Jul. 2025 - Sep. 2025**

AI Agentic Negotiation

- Extended a multi-agent negotiation evaluation framework, automated experiments, and reproducibility across local LLM models (e.g., LLaMA/Qwen families).
- Implemented belief updates and offer-counteroffer dynamics, designed metrics and dashboards for comparative analysis.

Wireless Systems and Applications Lab **Hanoi, Vietnam**
Research Assistant **Mar. 2019 - Mar. 2023**

Design and Security Analysis of Satellite-based Free-Space Quantum Key Distribution (QKD) Systems for Wireless and Vehicular Networks

- Designed and evaluated satellite-based QKD architectures, incorporating link-layer retransmissions, relaying schemes, and security performance metrics.
- Conducted MATLAB-based simulations to quantify performance and validate analytical results.
- **Outputs:** 2 published conference papers & 2 published journal papers.

EDUCATION

Oregon State University **Mar. 2022 - Sept. 2027 (Expected)**
Ph.D. in Electrical & Computer Engineering Corvallis, OR, US

Minor in Artificial Intelligence

Topics: Neural Data Compression, Machine Learning, Information Theory

Oregon State University **Mar. 2022 - Dec. 2024**
M.S. in Electrical & Computer Engineering Corvallis, OR, US

Thesis: *On Minimizing Symbol Error Probability using Beamforming in MIMO Wiretap Channels*

SELECTED PUBLICATIONS

2 journal articles, 7 conference articles. A detailed list of publications is available [here](#).

- [1] **Nam Nguyen**, Thinh Nguyen, and Bella Bose. *Perception-based Image Denoising via Generative Compression*. IEEE International Conference on Image Processing, 2026 (Oral Presentation). [\[PDF\]](#)
- [2] **Nam Nguyen**, Thinh Nguyen, and Bella Bose. *Cross-Domain Lossy Compression via Rate- and Classification-Constrained Optimal Transport*. International Conference on Learning Representations, 2026. (Oral Presentation). [\[PDF\]](#)
- [3] **Nam Nguyen**, Thinh Nguyen, and Bella Bose. *Rate-Distortion-Classification Representation Theory for Bernoulli Sources*. IEEE International Symposium on Information Theory, 2026 (Accepted). [\[PDF\]](#)
- [4] **Nam Nguyen**, Thuan Nguyen, Thinh Nguyen, and Bella Bose. *Universal Rate-Distortion-Classification Representations for Lossy Compression*. IEEE Information Theory Workshop, 2025. [\[PDF\]](#)

HONORS AND AWARD

- NSF iREDEFINE Fellow – ECE Department Heads Association 2026
- Signal Processing Society Scholarship – IEEE Signal Processing Society 2025
- NSF Student Travel Grant, AERPAP Spring Workshop – North Carolina State University 2025
- Graduate School's Scholarly Presentation Award – Oregon State University 2024, 2025
- Second Prize, National Scientific Research Contest – Vietnam 2020
- Second Prize in Physics, Provincial Excellent Student Competition – Vietnam 2012
- First Prize in Physics, School-Level Excellent Student Competition – Vietnam 2011, 2012, 2013